

James Motes

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EDUCATION AND TRAINING

Postdoctoral Researcher, University of Illinois Urbana Champaign Advisor: Dr. Nancy M. Amato	August 2023 - Present
Ph.D. in Computer Science, University of Illinois Urbana Champaign Ph.D. Thesis: Multi-robot Task and Motion Planning in Hybrid State Spaces Advisor: Dr. Nancy M. Amato	August 2023
M.S. in Computer Science, Texas A&M University M.S. Thesis: Interaction Templates for Multi-Robot Systems Advisor: Dr. Nancy M. Amato	August 2019
B.S. in Computer Engineering, Texas A&M University Minor in Mathematics, Engineering Honors, Undergraduate Research Scholar Undergraduate Thesis: Multi-Agent Persistent Task Performance Advisor: Dr. Nancy M. Amato	May 2018

RESEARCH INTERESTS

Multi-Robot Planning: Apply growing general intelligence within the complexity of multi-robot problems.

Human-Robot Co-planning: Develop co-planning systems in which humans and robots/AI jointly create and refine plans.

Hardware and Software Accelerated Planning: Design planning architectures that exploit modern HW/SW techniques.

Applications: Smart manufacturing and assembly; logistics and warehousing; agriculture; automated labs; healthcare and in-home assistive robotics

EXPERIENCE

Postdoctoral Researcher, Parasol Lab, University of Illinois Urbana Champaign - Lead multiple graduate research teams (17 graduate students across the lifetime of the projects, some carried over from Ph.D. Graduate Researcher position) - Primary author for grant proposals (Submissions to NSF Foundational Research in Robotics, 6 industry collaborations) - Lead software development for the open source deployment of the Parasol Planning Library (PPL) - Develop theoretical frameworks for multi-robot planning with applications in autonomous factories, logistics, biofabrication labs, and mining - Design interfaces for human-robot collaborative planning including natural language-conditioned motion planning and AR/VR interfaces for motion planning - Leverage computational geometry techniques and GPU serialization for accelerating core motion planning algorithms in dynamic environments	August 2023 - Present
Optigon Inc, Principal Software Engineer - Lead software development for DOE SBIR-funded, high-throughput, non-contact spectroscopic metrology and analysis tools for clean-energy materials (e.g., perovskite photovoltaics) - Advise on machine learning-based analytical models of solar cell behaviors	May 2024 - Present
Normandy Automation, Founder - Developed autonomous robotic manufacturing solutions - Leveraged AI and multi-robot systems for just-in-time manufacturing with minimal human reprogramming - Wrote NSF SBIR proposals for autonomous welding and CNC machine tending	May 2023 - August 2025
QuickReports, Founder - Developed software for healthcare management service organization business development teams - Provided autonomously generated competition and demographics reports for brick and mortar healthcare practices	May 2023 - August 2025

Lead Instructor, AI4All

August 2021 - December 2023

- Instructor for Discover AI introductory artificial intelligence courses at the University of Texas at El Paso, New Mexico State University (Spring 2022), Worcester Polytechnic Institute (Fall 2022 and Spring 2023), and Texas A&M University (Fall 2022, Spring 2023, Fall 2023)
- Developed course curriculum for Discover AI introductory artificial intelligence course (Summer 2022)
- Led and mentored team of instructors at other universities
- Teaching assistant for Discover AI introductory artificial intelligence course at the University of Illinois Urbana Champaign (Fall 2021)

Graduate Researcher, Parasol Lab, University of Illinois Urbana Champaign

August 2019 - August 2023

- Led graduate research team studying multi-robot planning
- Designed theoretical frameworks for large scale multi-robot task and motion planning with multiple publications in top robotics journals (IEEE T-RO and RA-L)
- Developed parallelized variants of core multi-agent pathfinding method
- Collaborated with Foxconn Interconnect Technologies and two other research labs at UIUC (Dr. Timothy Bretl, Dr. Katherine Driggs-Campbell) on a series of projects for a collaborative human-robot factory

Teaching Assistant, University of Illinois Urbana Champaign

January 2021 - May 2021

- Computer Science Ph.D. seminar course
- Organized guest speakers, managed grading and student assignments

Graduate Researcher, Parasol Lab, Texas A&M University

May 2018 - August 2019

- Developed the Interaction Template Method for reusing expensive precomputed coordinated interactions (e.g., handoffs) in multi-robot task and motion planning
- Culminated in master's thesis and IEEE RA-L publication
- Mentored undergraduate researchers who contributed to the publication

Undergraduate Research Scholar, Parasol Lab, Texas A&M University

September 2017 - May 2018

- Developed proactive multi-agent persistent task performance system under battery constraint
- Culminated in undergraduate thesis

Hewlett Packard Enterprise - Design Verification Engineer Intern

May - August 2017

- Worked on Gen-Z Switch Fabric project within the Silicon Design Lab
- Developed VLSI Design Verification components utilizing Universal Verification Methodology (UVM)

HONORS

Engineering Graduate Merit Fellowship	August 2018 - August 2019
Presidential Endowed Scholarship	August 2014 - May 2018
Distinguished Student Award - Dwight Look College of Engineering	May 2017
Industrial Affiliates Program Scholarship	August 2014 - May 2018
Dell Merit Scholarship	August 2017 - May 2018
Salutatorian, Rowlett High School (2/591)	June 2014

PUBLICATIONS (12, Google Scholar citations = 312, h = 7 as of 11/30/2025) **direct mentorship**Journal Articles (9)**

- Lee, H.,** **Motes, J. D.**, Morales, M., & Amato, N. M., An Analysis of Constraint-Based Multi-Agent Pathfinding Algorithms. (2026). *IEEE Transactions on Robotics*, to appear.
- Qin, M.,** Solis, I.,** **Motes, J. D.**, Morales, M., & Amato, N. M. (2025). K-ARC: Adaptive Robot Coordination for Multi-Robot Kinodynamic Planning. *IEEE Robotics and Automation Letters*. To be presented at the International Conference on Robotics and Automation (ICRA), Vienna, Austria, June, 2026.
- Solis, I.,** **Motes, J.**, Qin, M.,** Morales, M., & Amato, N. M. (2024). Adaptive Robot Coordination: A Subproblem-based Approach for Hybrid Multi-robot Motion Planning. *IEEE Robotics and Automation Letters*. Presented at ICRA@40, Rotterdam, Netherlands, September, 2024.
- McBeth, C.,** **Motes, J.**, Uwacu, D., Morales, M., & Amato, N. M. (2023). Scalable Multi-robot Motion Planning for Congested Environments With Topological Guidance. *IEEE Robotics and Automation Letters*. Presented at the International Conference on Robotics and Automation (ICRA), Yokohama, Japan, May, 2024.
- **Motes, J.**, Chen, T., Bretl, T., Aguirre, M. M., & Amato, N. M. (2023). Hypergraph-based multi-robot task and motion planning. *IEEE Transactions on Robotics*. Presented at the International Conference on Robotics and Automation (ICRA), Yokohama, Japan, May, 2024.

- Lee, H.,** **Motes, J.**, Morales, M., and Amato, N. M., 2021. Parallel Hierarchical Composition Conflict-Based Search for Optimal Multi-Agent Pathfinding. *IEEE Robotics and Automation Letters*, 6(4), pp. 7001-7008. Presented at the International Conference on Intelligent Robots and Systems (IROS), Prague, Czech Republic, 2021 (virtual).
- Solis, I.,** **Motes, J.**, Sandström, R., and Amato, N. M., 2021. Roadmap-Optimal Multi-Robot Motion Planning using Conflict-Based Search. *IEEE Robotics and Automation Letters*, 6(3), pp. 4608-4615. Presented at the International Conference on Robotics and Automation (ICRA), Xi'an, China, 2021 (virtual).
- **Motes, J.**, Sandström, R., Lee, H.,** Thomas, S., and Amato, N. M., April 2020. "Multi-Robot Task and Motion Planning With Subtask Dependencies," in *IEEE Robotics and Automation Letters*, 5(2), pp. 3338-3345. Presented at the International Conference on Robotics and Automation, Paris (ICRA), France, 2020 (virtual).
- **Motes, J.**, Sandström, R., Adams, W., Ogunyale, T.,** Thomas, S. and Amato, N.M., 2019. Interaction Templates for Multi-Robot Systems. *IEEE Robotics and Automation Letters*, 4(3), pp. 2926-2933. Presented at the International Conference on Intelligent Robots and Systems (IROS), Macao, China, 2019.

Refereed Conference Proceedings (2)

- Goddu, A., Brown, T., Paton, M., **Motes, J.**, & Chen, T. (2025). Modifying ABIT* for Tethered Rappelling Robot Motion Planning. *2025 IEEE 21st International Conference on Automation Science and Engineering (CASE)* (pp. 722-728). IEEE, 2025.
- Ashur, S., Lusardi, M., Markowicz, M., **Motes, J.**, Morales, M., Har-Peled, S., & Amato, N. M. (2024). SPITE: Simple Polyhedral Intersection Techniques for Modified Environments. *Algorithmic Foundations of Robotics XVI, Proceedings of the Sixteenth Workshop on the Algorithmic Foundations of Robotics*, to appear.

Workshop Papers (1)

- Chen, T., Huang, Z., **Motes, J.**, Geng, J., Ta, Q., Dinkel, H., Abdul-Rashid, H., Myers, J., Mun, Y., Lin, W., Huang, Y., Liu, S., Morales, M., Amato, N. M., Driggs-Campbell, K., Bretl, T., 2022. Insights from an Industrial Collaborative Assembly Project: Lessons in Research and Collaboration. *IEEE ICRA Workshop on Collaborative Robots and the Work of the Future*, Philadelphia, USA, 2022.

UNDER REVIEW (6 as of 11/30/2025)

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- Ngui, I., McBeth, C., Santos, A., He, G., Mimnaugh, K. J., **Motes, J. D.**, Soares, L., Morales, M. and Amato, N.M., ERUPT: An Open Toolkit for Interfacing with Robot Motion Planners in Extended Reality. *arXiv preprint arXiv:2510.02464*.
 - Lee, S.,** **Motes, J. D.**, Ngui, I.,** Morales, M., & Amato, N. M., Lazy-DaSH: Lazy Approach for Hypergraph-based Multi-robot Task and Motion Planning. *arXiv preprint arXiv:2504.05552*.
 - Lee, H.,** Serlin, Z., **Motes, J. D.**, Long, B., Morales, M., & Amato, N. M., PRISM: Complete Online Decentralized Multi-Agent Pathfinding with Rapid Information Sharing using Motion Constraints. *arXiv preprint arXiv:2505.08025*.
 - Solis, I.,** **Motes, J. D.**, Qin, M.,** Morales, M., & Amato, N. M., Experience-based Subproblem Planning for Multi-Robot Motion Planning. *arXiv preprint arXiv:2411.08851*.
 - McBeth, C.,** **Motes, J. D.**, Ngui, I.,** Morales, M., & Amato, N. M., Scalable Multi-Robot Motion Planning Using Guidance-Informed Hypergraphs. *arXiv preprint arXiv:2311.10176*.
 - Attali, A., Ashur, S., Love, I. B.,** McBeth, C.,** **Motes, J. D.**, Uwacu, D., Morales, M. and Amato, N.M., A Framework for Guided Motion Planning. *arXiv preprint arXiv:2404.03133*.

IN PREPARATION FOR 2025 (5 as of 11/30/2025)

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- **Motes, J. D.**, Markowicz, M.,** Ashur, S., Arad, Y.,** Morales, M., & Amato, N. M., sLRS: semi-Lazy Rearrangement Solver - Leveraging Dynamic Roadmap Graphs in Task and Motion Planning.
 - Arad, Y.,** Markowicz, M.,** Ashur, S., **Motes, J. D.**, Morales, M., & Amato, N. M., Serialized Red-Green-Gray: A Quick Heuristic Validation of Edges in Dynamic Roadmap Graphs.
 - Markowicz, M.,** Arad, Y.,** Ashur, S., **Motes, J. D.**, Morales, M., & Amato, N. M., Greedy Lazy SPITE: Heuristic Validation and Lazy Motion Planning with Dynamic Roadmap Graphs.
 - Markowicz, M.,** **Motes, J. D.**, Solis, I.,** Morales, M., & Amato, N. M., MR-SPITE: Leveraging Dynamic Roadmap Graphs in Multi-robot Motion Planning.
 - Love, I. B.,** Lohan, D. J., **Motes, J. D.**, Dede, E. M., Morales, M., & Amato, N. M., A Hierarchical Approach for the Generation of Varied Cable Harness Designs.

NON-PEER REVIEWED (8 as of 11/30/2025)

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- Markowicz, M.,** Ashur, S.,** **Motes, J.**, and Amato, N. M. "Semi-Lazy Rearrangement Solver for Accelerating Task Planning Using SPITE Dynamic Roadmaps." Extended Abstract for *4th Workshop on Future of Construction: Safe, Reliable, and Precise Robots in Construction Environments, ICRA 2025*, Georgia World Congress Center, Atlanta, Georgia, USA, May 19, 2025.

- Lee, H.,** **Motes, J.**, Serlin, Z., Morales, M., and Amato, N. "Distributed Constraint-Based Search using Multi-Hop Communication." Extended abstract for *ICRA@40: 40th Anniversary of the IEEE Conference on Robotics and Automation (ICRA@40)*, Rotterdam, The Netherlands, Sept. 23–26, 2024.
- Ngui, I.,** **Motes, J.**, Lee, H.,** Morales, M., and Amato, N. M. "Ergonomic Motion Planning for Human-Robot Collaboration." Extended abstract for *ICRA@40: 40th Anniversary of the IEEE Conference on Robotics and Automation (ICRA@40)*, Rotterdam, The Netherlands, Sept. 23–26, 2024.
- Elimelech, K.* **Motes, J.***, Morales, M., Amato, N. M., Vardi, M., and Kavraki, L. E. "Encoding Reusable Multi-Robot Planning Strategies as Abstract Hypergraphs." Extended abstract for *ICRA@40: 40th Anniversary of the IEEE Conference on Robotics and Automation (ICRA@40)*, Rotterdam, The Netherlands, Sept. 23–26, 2024. *Equal contribution.
- Lee, S.** **Motes, J.**, Ngui, I.,** Morales, M., and Amato, N. M. "Lazy-DaSH: Lazy Approach for Hypergraph-based Multi-robot Task and Motion Planning." Extended abstract for *ICRA@40: 40th Anniversary of the IEEE Conference on Robotics and Automation (ICRA@40)*, Rotterdam, The Netherlands, Sept. 23–26, 2024.
- Lusardi, M., Markowicz, M., Ashur, S., **Motes, J.**, Morales, M., Har-Peled, S., and Amato, N. M. "SPITE: Simple Polyhedral Intersection Techniques for Modified Environments." Extended abstract for *ICRA@40: 40th Anniversary of the IEEE Conference on Robotics and Automation (ICRA@40)*, Rotterdam, The Netherlands, Sept. 23–26, 2024.
- Ngui, I.,** Lee, S.,** **Motes, J.**, Morales, M., and Amato, N. M. "A Hierarchical Approach to Workstation-based Task-Allocation and Motion Planning." Extended Abstract for *RAFF2023: Robotics & AI in Future Factory (IROS 2023 Workshop)*, IROS 2023, Detroit, Michigan, USA, Oct. 1–5, 2023.
- Attali, A., Ashur, S., Love, I. B., McBeth, C., **Motes, J.**, Uwacu, D., Morales, M., and Amato, N. M. "Evaluating guiding spaces for motion planning." Extended Abstract for *Evaluating Motion Planning Performance, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2022)*, Room 12/J, Kyoto, Japan, Oct. 23, 2022.

FUNDING AUTHORSHIP

National Science Foundation (NSF) - Foundational Research in Robotics (FRR) Proposal September 2025

- Primary author on proposal for customizing robotic motion behaviors for non-expert users via natural language preference specification
- Collaboration with Natural Language Processing researchers at University of Illinois Urbana Champaign

Industry Proposals and Whitepapers April 2025 - Present

- Primary author on six proposals (secondary author on three more) covering human-robot interaction, multi-robot planning, environment design, and leveraging data in planning

TALKS

- **February 2024 - Invited Lab Seminar.** "Multi-Robot Task and Motion Planning in Hybrid State Spaces." *Kavraki Lab, Department of Computer Science, Rice University, Houston, TX*
- **October 2019 - Invited Workshop Talk.** "Planning Motions and Tasks for Manipulators, Multi-Robot Systems and Biomolecules." *Center for Autonomy Portfolio Discovery Workshop, UIUC, Urbana, IL*

LEADERSHIP, MENTORING, AND PROFESSIONAL ACTIVITIES

Postdoctoral Researcher, University of Illinois Urbana Champaign (UIUC) August 2023 - Present

- Mentor 17 graduate students (5 graduated), some carried over from Ph.D. Graduate Researcher
- Led to 5 published manuscripts, 7 under review, and 5 in preparation for submission in 2025 (15 total)

Houston Robotics Club August 2023 - Present

- Mentor community college and high school students on AI and robotics projects
- Several now enrolled in STEM undergraduate programs or at internships with local robotics companies

Graduate Researcher, UIUC and Texas A&M University May 2018 - August 2023

- Mentored graduate students (14) within my research group
- Mentored numerous undergraduate students, both local university students and DREU program students
- Led to 7 published manuscripts
- Taught motion planning course for new lab members (2019-2021)

Program Committee Member

- WAFR 2026

- WAFR 2024

Member of IEEE Robotics and Automation Society

May 2019 - Present

Reviewer/Referee

August 2018 - Present

- IEEE Robotics and Automation Letters (RA-L)
- IEEE Transactions on Robotics (T-RO)
- IEEE International Conference on Robotics and Automation (ICRA)
- IEEE International Conference on Intelligent Robots and Systems (IROS)
- IEEE International Symposium on Multi-robot and Multi-agent Systems (MRS)
- World Symposium (formerly International Workshop) on Algorithmic Foundations of Robotics (WAFR)
- International Journal of Robotics Research (IJRR)